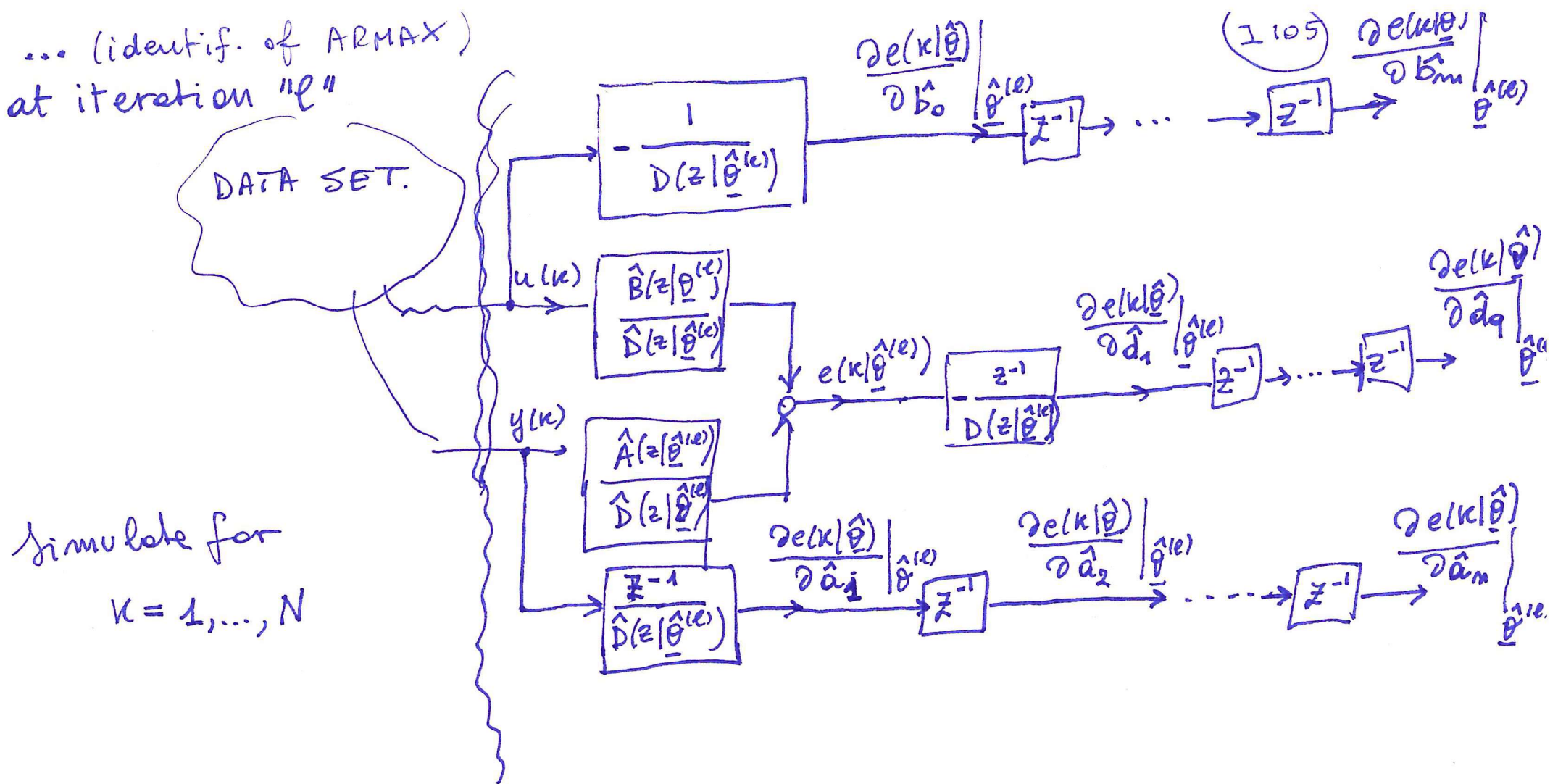


... (identif. of ARMAX)
at iteration "l"



$$\Rightarrow \underline{\hat{\theta}}^{(l+1)} = \underline{\hat{\theta}}^{(l)} - \alpha^{(l)} [\dots]$$

until convergence (verification of the stopping criterion.)

Application of the same alg. to Output Error.

$$\hat{y}(k|\underline{\hat{\theta}}) = \frac{\hat{B}(z|\underline{\hat{\theta}})}{\hat{A}(z|\underline{\hat{\theta}})} u(k)$$

$$e(k|\underline{\hat{\theta}}) = y(k) - \frac{\hat{B}(z|\underline{\hat{\theta}})}{\hat{A}(z|\underline{\hat{\theta}})} u(k) \iff \hat{A}(z|\underline{\hat{\theta}}) e(k|\underline{\hat{\theta}}) = \hat{A}(z|\underline{\hat{\theta}}) y(k) - \hat{B}(z|\underline{\hat{\theta}}) u(k)$$

$$\vdots \quad \frac{\partial}{\partial a_i} \quad i=1 \dots n \quad \frac{\partial}{\partial b_i} \quad i=0 \dots m$$

(do by exercise)

ARX:

$$\hat{y}(k|\underline{\hat{\theta}}) = [1 - \hat{A}(z|\underline{\hat{\theta}})] y(k) + \hat{B}(z|\underline{\hat{\theta}}) u(k)$$

$$= \underline{\psi}^T(k) \underline{\hat{\theta}}$$

one shot identif.
(batch mode)
"of line"
with all the data set
 $\psi(1) \dots \psi(N)$

$$\Rightarrow \underline{\hat{\theta}}^* = \arg \min_{\underline{\hat{\theta}}} J_N(\underline{\hat{\theta}}) = (\Phi_N^T \Phi_N)^{-1} \Phi_N^T y_N$$

where $J_N(\underline{\hat{\theta}}) = \frac{1}{N} \sum_{k=1}^N e^2(k|\underline{\hat{\theta}})$

$$e(k|\underline{\hat{\theta}}) = y(k) - \hat{y}(k|\underline{\hat{\theta}})$$

This computation can be performed "off line"

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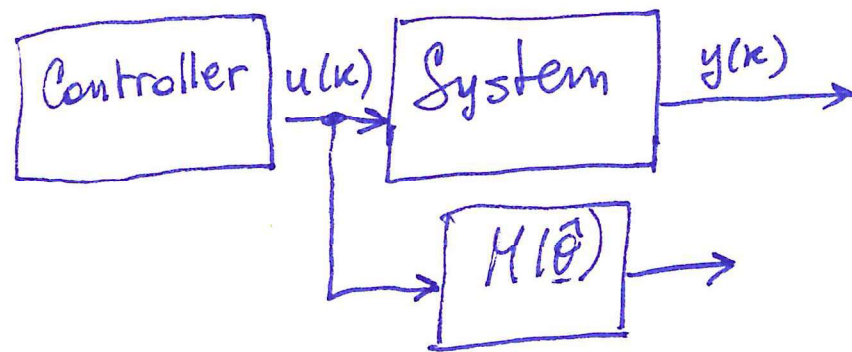
It can be interpreted as the identification after N -steps.

we could denote by $\underline{\hat{\theta}}^*(N) = (\underline{\Phi}_N \underline{\Phi}_N^T)^{-1} \underline{\Phi}_N^T y_N$

We could want to estimate $\underline{\theta}$ while the system is running (ON LINE)
on-line identification of the parameters.

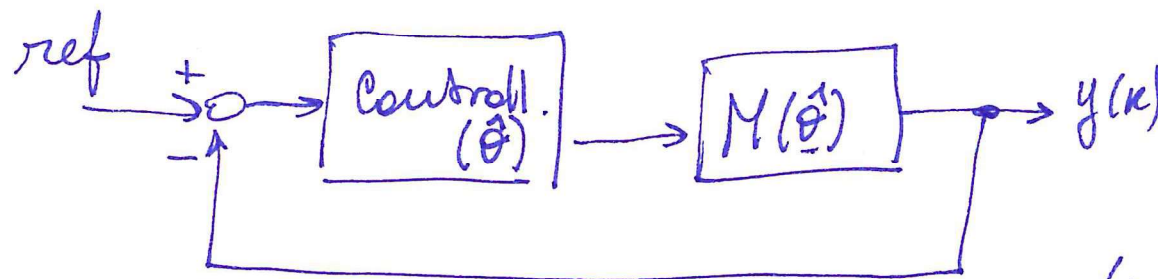
Useful in an "adaptive control" framework

$M(\underline{\hat{\theta}})$ = model for the system.



If $M(\underline{\hat{\theta}}) \equiv \text{System}$

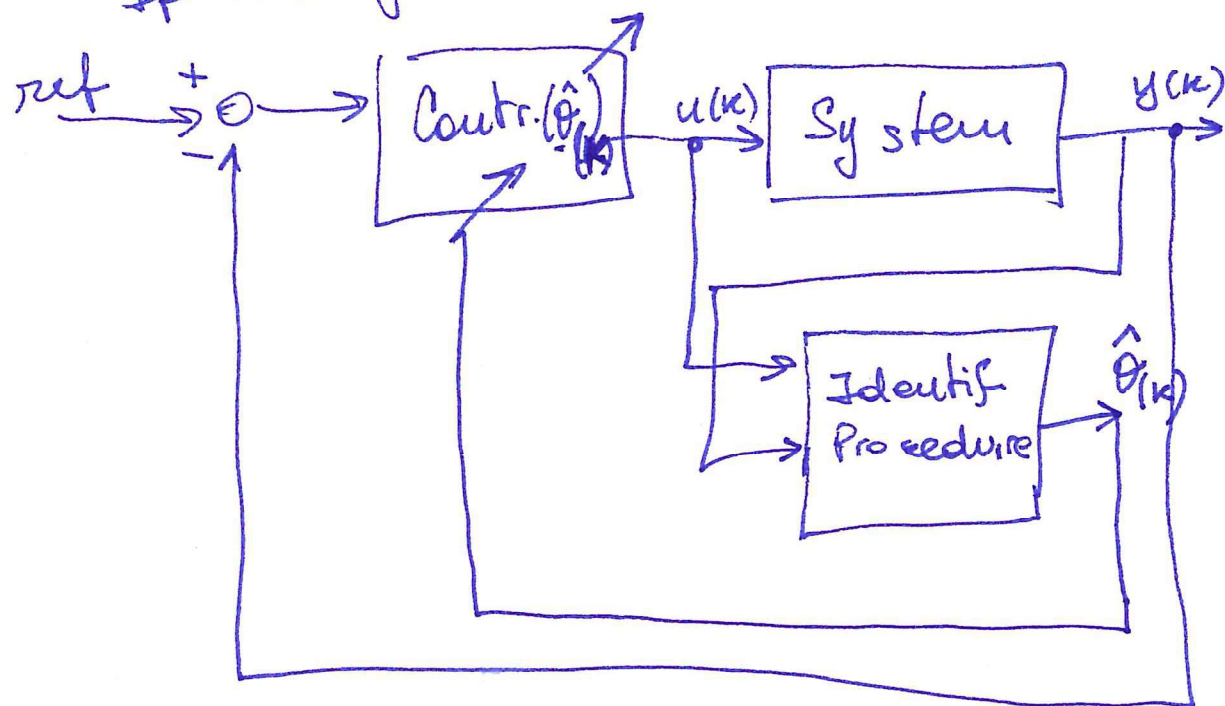
$\Downarrow$



Controller could be
"parametrized by $\underline{\hat{\theta}}$ "

(i.e. designed using $\underline{\hat{\theta}}$)

If the system is unknown.



Adaptive Control

based on

Certainty of Equivalence.

"As is $H(\underline{\theta})$ true"

$\Rightarrow$ we need an estimate $\underline{\hat{\theta}}(k)$ $k = 0, 1, \dots$

and a method for estimating $\underline{\theta}$ at time k in a way not too computationally heavy.

In principle: $\underline{\hat{\theta}}^*(k) = (\underline{\Phi}_k \underline{\Phi}_k^T)^{-1} \underline{\Phi}_k^T y_k$

where $\underline{\Phi}_k \triangleq [\varphi(1) | \dots | \varphi(k)]$

$y_k \triangleq \begin{bmatrix} y(1) \\ \vdots \\ y(k) \end{bmatrix}$

this corresp. to using K ~~data~~ instead of N

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i.e.

$$\hat{\underline{\theta}}^*(k) = \arg \min_{\hat{\underline{\theta}}} J_k(\hat{\underline{\theta}}) = \arg \min_{\hat{\underline{\theta}}} \underbrace{\frac{1}{K} \sum_{i=1}^K e^2(i | \hat{\underline{\theta}})}_{\text{least squares cost for the first } K \text{ steps.}}$$

$$\Phi_k = [\varphi(1) | \dots | \varphi(k)] \quad p \times K \quad p = \dim \hat{\underline{\theta}} = n+m+1$$

$$\begin{matrix} (\Phi_k \Phi_k^T)^{-1} \\ \begin{matrix} \downarrow & \downarrow \\ p \times k & k \times p \end{matrix} \end{matrix} \quad p \times p$$

of columns of Φ_k grows with $k \Rightarrow$ we cannot apply the one-shot alg. for growing k .

We want to resort to a RECURSIVE ARX method. using at each step a very limited computational load.

Maybe the system slowly changes while running.

(I no)

$\Rightarrow$ forgetting factor: informations far in the past must influence the estimate "not so much"

$$\Rightarrow J_k(\hat{\theta}) = \frac{1}{k} \sum_{i=1}^k \mu^{k-i} e^2(i|\hat{\theta})$$

$$0 < \mu < 1$$

$\mu = 1 \Rightarrow$ No forgetting
(as before)

generalization of $\hat{\theta}^*(k) = \arg \min_{\hat{\theta}} J_k(\hat{\theta})$

for $\mu \neq 1$

$$\mu = 1 \Rightarrow \hat{\theta}^*(k) = (\Phi_k \Phi_k^T)^{-1} \Phi_k^T y_k$$

$$J_k(\hat{\theta}) = \frac{1}{k} \underline{e}_k^T(\hat{\theta}) \underline{e}_k(\hat{\theta})$$

We can rewrite $J_k(\hat{\theta}) = \frac{1}{k} \underline{e}_k^T(\hat{\theta}) W_k \underline{e}_k(\hat{\theta})$

$$\underline{e}_k(\hat{\theta}) = \begin{bmatrix} e(1|\hat{\theta}) \\ \vdots \\ e(k|\hat{\theta}) \end{bmatrix}$$

$$\text{where } W_k = \begin{bmatrix} \mu^{k-1} & & \\ & \mu^{k-2} & \\ & & \ddots \\ \phi & & & \mu^{k-k} = 1 \end{bmatrix}$$

in fact

$$\underline{e}_k^T(\hat{\underline{\theta}}) W_k \underline{e}_k(\hat{\underline{\theta}}) = [e(1|\hat{\underline{\theta}}) \dots e(k|\hat{\underline{\theta}})] \begin{bmatrix} \mu^{k-1} & & \\ & \ddots & \\ & & \mu^{k-k} \end{bmatrix} \begin{bmatrix} e(1|\hat{\underline{\theta}}) \\ \vdots \\ e(k|\hat{\underline{\theta}}) \end{bmatrix}$$

$$= e(1|\hat{\underline{\theta}}) \mu^{k-1} e(1|\hat{\underline{\theta}}) + \dots + e(k|\hat{\underline{\theta}}) \mu^{k-k} e(k|\hat{\underline{\theta}}) =$$

$$= \sum_{i=1}^k \mu^{k-i} e^2(i|\hat{\underline{\theta}})$$

with very "same" computations as in the case

$$W_k = I \quad (k=1)$$

$$\Rightarrow J_k(\hat{\underline{\theta}}) = \frac{1}{k} \underline{e}_k^T(\hat{\underline{\theta}}) W_k \underline{e}_k(\hat{\underline{\theta}}) = \dots = \frac{1}{k} \underline{y}_k^T W_k \underline{y}_k +$$

$$+ \frac{2}{k} \left[\frac{1}{2} \hat{\underline{\theta}}^T \Phi_k W_k \Phi_k^T \hat{\underline{\theta}} - \underline{y}_k^T W_k \Phi_k^T \hat{\underline{\theta}} \right]$$

$$\Rightarrow (\dots) \left[\Phi_k W_k \Phi_k^T \right] \hat{\underline{\theta}}^*(k) = \Phi_k W_k \underline{y}_k \quad (\text{Normal equation})$$

and if $[\cdot]$ is invertible
one shot.

$$\Rightarrow \hat{\underline{\theta}}^*(k) = \left(\Phi_k W_k \Phi_k^T \right)^{-1} \Phi_k W_k \underline{y}_k$$

$$S_k \triangleq \Phi_k W_k \Phi_k^T$$

$$Q_k \triangleq \bigoplus_k W_k y_k$$

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$$S_K = [\varphi(1) | \dots | \varphi(K)] \begin{bmatrix} \mu^{K-1} & & \\ & \ddots & \\ & & \mu^{K-K} \end{bmatrix} \begin{bmatrix} \varphi^+(1) \\ \vdots \\ \varphi^+(K) \end{bmatrix} =$$

$$= [\psi(1) \mid \dots \mid \psi(k)] \cdot \begin{bmatrix} \mu^{k-1} \psi^T(1) \\ \vdots \\ \mu^{k-k} \psi^T(k) \end{bmatrix}$$

In general:

$$\left[\underline{a}_1 \mid \dots \mid \underline{a}_k \right] \begin{bmatrix} \underline{b}_1^T \\ \vdots \\ \underline{b}_k^T \end{bmatrix} = \sum_{i=1}^k \underline{a}_i \underline{b}_i^T$$

In our case $S_K = \sum_{i=1}^K \varphi(i) \mu^{K-i} \varphi^T(i) = \sum_{i=1}^K \mu^{K-i} \varphi(i) \varphi^T(i) \in \mathbb{R}^{p \times p}$

$$= 1 \cdot \underline{\varphi(k)} \underline{\varphi(k)}^T + \sum_{i=1}^{k-1} \mu^{k-i} \underline{\varphi(i)} \underline{\varphi(i)}^T = (\dots)$$

$$S_k = (\dots) = \varphi(k) \varphi^T(k) + \underbrace{\mu \sum_{i=1}^{k-1} \mu^{(k-1)-i} \varphi(i) \varphi^T(i)}_{\triangleq S_{k-1}} =$$

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summing up:

$$S_k = \varphi(k) \varphi^T(k) + \mu S_{k-1}$$

but also: $S_{k-1} = \frac{1}{\mu} [S_k - \varphi(k) \varphi^T(k)]$

$$Q_k = \Phi_k W_k y_k = [\varphi(1) \dots \varphi(k)] \cdot \begin{bmatrix} \mu^{k-1} y(1) \\ \vdots \\ \mu^{k-k} y(k) \end{bmatrix} = \sum_{i=1}^k \varphi(i) \mu^{k-i} y(i) =$$

$$= \varphi(k) y(k) + \mu \sum_{i=1}^{k-1} \mu^{(k-1)-i} \varphi(i) y(i) = \varphi(k) y(k) + \mu Q_{k-1}$$

Recall: $S_k \hat{\vartheta}(k) = Q_k$

for $k-1 \Rightarrow Q_{k-1} = S_{k-1} \hat{\vartheta}^*(k-1)$

We want to compute

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$$\begin{aligned}
 \hat{\underline{\theta}}^*(k) &= S_k^{-1} Q_k = S_k^{-1} \left[\underline{\varphi}(k) y(k) + \mu Q_{k-1} \right] = \\
 &= S_k^{-1} \left[\underline{\varphi}(k) y(k) + \mu S_{k-1} \hat{\underline{\theta}}^*(k-1) \right] = \\
 &= S_k^{-1} \left[\underline{\varphi}(k) y(k) + \underbrace{\mu \frac{1}{\mu} (S_k - \underline{\varphi}(k) \underline{\varphi}^T(k))}_{S_{k-1}} \hat{\underline{\theta}}^*(k-1) \right] = \\
 &= \underbrace{S_k^{-1} \underline{\varphi}(k) y(k)} + \underbrace{S_k^{-1} S_k}_{I} \hat{\underline{\theta}}^*(k-1) - \underbrace{S_k^{-1} \underline{\varphi}(k) \underline{\varphi}^T(k)} \hat{\underline{\theta}}^*(k-1) \\
 \dots = \quad \hat{\underline{\theta}}^*(k) &= \hat{\underline{\theta}}^*(k-1) + \underbrace{S_k^{-1} \underline{\varphi}(k)}_{\parallel} \left[y(k) - \underline{\varphi}^T(k) \hat{\underline{\theta}}^*(k-1) \right]
 \end{aligned}$$

OR:

$$\boxed{ \hat{\underline{\theta}}^*(k) = \hat{\underline{\theta}}^*(k-1) + \underline{\chi}(k) \left[y(k) - \hat{y}(k | \hat{\underline{\theta}}^*(k-1)) \right] }$$

↑
new value

↑
predicted value
based on $\hat{\underline{\theta}}^*(k-1)$